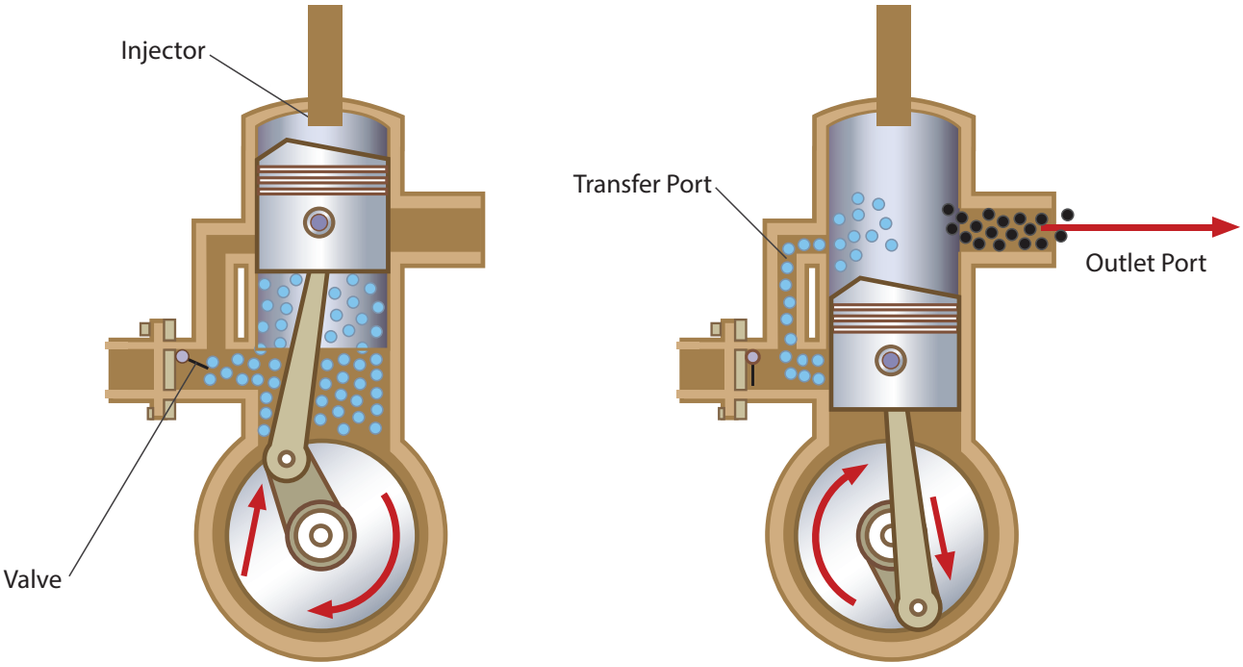


Fig 5.8(b) Upward and Downward Stroke of Two Stroke Diesel Engine

Further, the upward movement of the piston a partial vacuum is created in the crankcase and this allows the entry of the fresh charge into the crankcase through uncovered inlet port. The exhaust port and the inlet port remains covered when the piston at the TDC.

Downward stroke:

As soon as the combustion of the fresh charge takes place, a large amount of the hot gases is produced and this exerts a very high pressure force on the top of the piston. Due to this high pressure force, the piston moves downward and rotates the crankshaft and does useful work.

During this stroke the inlet port is covered by the piston and the new charge is compressed in the crankcase due to the downward movement of piston.

Further downward movement of the piston uncovers first the exhaust and the exhaust starts through the exhaust port.

Further downward movement of the piston uncovers port the transfer port and the fresh air is forced into the cylinder.

The fresh air strikes the deflector on the piston crown, rises to the top of the cylinder and pushes out most of the exhaust gases.

The piston is now at BDC position. The cylinder is completely filled with the fresh charge but it is somewhat diluted with the exhaust gases.

Finally the cycle event is then repeated and the power stroke is obtained for the every single revolution of the crankshaft.

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